

CreditCardPayment.java

Teacher's Guide & Speaker Notes for Presentation

TEACHING THE CLASS: Inheritance

```
public class CreditCardPayment extends Payment
```

How to explain this: Point out the word "extends". Tell your audience this is Inheritance! Explain that because of this one word, we didn't have to write the code for Transaction IDs or Timestamps in this file. We got all of that for free from the parent class.

TEACHING THE VARIABLES: Data Isolation

```
private String cardNumber;  
private String cvv;
```

How to explain this: Highlight that these are "private". This is a great time to talk about security. Explain that the PayPal class and the PaymentProcessor have absolutely no idea what the credit card number is. The data is locked entirely inside this specific class.

TEACHING THE ALGORITHM: Overriding

```
@Override  
public boolean process(...)
```

How to explain this: Point out "@Override". Explain that this is where we fulfill the contract the parent class demanded.

Walk them through the logic:

1. Show how we validate the card by stripping spaces and checking if the CVV is exactly 3 digits using a Regex string.
2. Show how we simulate a bank network by checking if the amount is under \$500,000.
3. Explain that if it succeeds, this child class reaches up into the parent class and changes the inherited status variable to "SUCCESS".